Residual Stream Geometry Tracks Task Structure, Not Context Length: Evidence from Game-Playing Transformers

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Abstract

We investigate whether residual stream geometry in transformers simplifies as models process more in-context examples during game-playing tasks—the hypothesized "many-shot effect." We train small GPT-style transformers (208K parameters) on sequences from a 3-state hidden Markov model (MESS3), Rock-Paper-Scissors with three opponent strategies, and TIC-TAC-TOE, then measure effective dimensionality (participation ratio), PCA concentration, and linear probe accuracy as functions of context position. Our key finding is that geometric simplification is driven primarily by *state-space narrowing* in the task itself, not by a universal many-shot learning mechanism. TIC-TAC-TOE exhibits dramatic dimensionality collapse (participation ratio: $7.85 \rightarrow 1.26$, Cohen's $d = 19.98$) because later moves are increasingly constrained. In contrast, stationary processes show approximately flat geometry across context, with participation ratio stabilizing around 2.0–2.3 after the first few tokens. Unexpectedly, belief state regression quality *decreases* with context position in MESS3 (R^2: $1.0 \rightarrow 0.95$, Spearman $\rho = -0.78$, $p < 0.001$), suggesting representational drift rather than refinement. These results demonstrate that residual stream dimensionality reflects the effective entropy of the data-generating process at each position, not the accumulation of in-context evidence.

1 Introduction

Transformers that process many in-context examples narrow their output distributions: with enough examples of a pattern, the model becomes more "certain" about what comes next. A natural question follows: **does this distributional narrowing correspond to geometric simplification in the residual stream?** If so, residual stream geometry could serve as a mechanistic signature of in-context learning progress, with applications to interpretability, uncertainty quantification, and understanding the limits of context utilization.

Recent work in mechanistic interpretability has established that transformers linearly encode belief state geometry in their residual streams Shai et al. [2024], and that game-playing models develop emergent world representations accessible via linear probes Nanda et al. [2023], Li et al. [2023]. However, these studies analyze geometry at fixed context positions without tracking how it evolves as context grows. The many-shot regime—where models process tens or hundreds of in-context examples—remains geometrically uncharacterized.

We address this gap by training small decoder-only transformers on sequences from games of varying complexity: a 3-state hidden Markov model (MESS3), Rock-Paper-Scissors with counter, random, and win-stay-lose-shift strategies, and TIC-TAC-TOE. For each model, we extract residual stream activations at every context position and measure three geometric properties: effective dimensionality via participation ratio, PCA variance concentration, and linear probe accuracy for latent game states.

Our main finding challenges the many-shot simplification hypothesis. Geometric simplification is not a universal consequence of processing more context. Instead, it tracks the *effective entropy* of the data-generating process at each position:

- TIC-TAC-TOE shows dramatic dimensionality collapse (PR: $7.85 \rightarrow 1.26$, Cohen's $d = 19.98$) because the game's state space shrinks as the board fills—a structural property of the task, not a learning phenomenon.
- Stationary processes (MESS3, RPS) show approximately constant geometry across context ($\mathrm{PR} \approx 2.0$–2.3), stabilizing within the first $\sim$5 positions regardless of how much additional context is provided.
- Belief state regression quality unexpectedly *decreases* with context position in MESS3 (R^2: $1.0 \rightarrow 0.95$, $p < 0.001$), revealing representational drift rather than progressive refinement.

Contributions. We make the following contributions:

- We provide the first systematic measurement of residual stream geometry as a function of context position across multiple game-playing tasks, testing the many-shot geometric simplification hypothesis.
- We demonstrate that effective dimensionality tracks task-level state-space entropy rather than context accumulation, with $\mathrm{PR} \approx 2$ emerging as a consistent floor for 3-token vocabularies—consistent with the belief state simplex being 2-dimensional.
- We identify an unexpected belief state regression degradation with context position, suggesting that small transformers experience representational drift over long sequences.

The remainder of this paper is organized as follows. Section 2 reviews related work on residual stream geometry and in-context learning. Section 3 describes our experimental setup, including data generation, model architecture, and geometric metrics. Section 4 presents our findings across all six datasets. Section 5 interprets the results and discusses limitations. Section 6 summarizes our contributions and outlines future directions.

2 Related Work

Belief state geometry in transformers. Shai et al. Shai et al. [2024] established that transformers trained on next-token prediction linearly represent the geometry of Bayesian belief states in their residual streams. Working with hidden Markov models including the MESS3 process, they showed that belief state simplices—including those with fractal structure—are faithfully embedded via affine maps $\boldsymbol{b} \approx \boldsymbol{W}\boldsymbol{a} + \boldsymbol{c}$, where $\boldsymbol{b}$ is the belief state and $\boldsymbol{a}$ is the residual stream activation. Piotrowski et al. Piotrowski et al. [2025] extended this framework, showing that transformers implement "constrained Bayesian belief updating" through attention heads that perform scalar updates in orthogonal spectral modes. Both studies analyze geometry at fixed context lengths; we extend this line of work by measuring how geometry evolves across context positions.

World models in game-playing transformers. Li et al. Li et al. [2023] demonstrated that a GPT model trained on Othello move sequences develops an internal world model of the board state, recoverable via nonlinear probes. Nanda et al. Nanda et al. [2023] refined this finding, showing that OthelloGPT encodes board state *linearly* when the representation is relative to the current player (mine/yours/empty rather than black/white/empty). He et al. He et al. [2024] applied sparse autoencoders to discover interpretable circuits in OthelloGPT without patch-based methods. These works establish that game-playing transformers develop structured internal representations, but do not track how representation geometry changes across the game.

In-context learning mechanisms. Olsson et al. Olsson et al. [2022] identified induction heads as a key mechanism for in-context learning, showing a phase transition during training where models acquire the ability to match and complete patterns from context. Garg et al. Garg et al. [2022] showed that transformers can implicitly implement gradient descent on in-context examples for simple function classes. These works characterize *how* in-context learning occurs but do not address the geometric consequences in the residual stream as context grows.

Geometric structure of representations. Engels et al. Engels et al. [2024] showed that some features in language models are represented as multi-dimensional subspaces rather than single directions, identifying "irreducible" multi-dimensional features. This is relevant to our work because

Table 1: Dataset summary. All datasets use 10,000 sequences with a 90/10 train/validation split.

Dataset	Seq. Length	Vocab Size	Entropy	Structure
MESS3 (p=0.9)	128	3	$\log 3 = 1.10$	Stationary HMM
MESS3 (p=0.6)	128	3	$\log 3 = 1.10$	Stationary HMM
RPS-COUNTER	128	3	$\log 3 = 1.10$	Deterministic strategy
RPS-RANDOM	128	3	$\log 3 = 1.10$	No learnable pattern
RPS-WSLS	128	3	$\log 3 = 1.10$	Conditional strategy
TIC-TAC-TOE	9	10	$\log 9 = 2.20$	Shrinking state space

belief states over k-state processes naturally inhabit $(k-1)$-dimensional simplices, making multi-dimensional representations theoretically necessary. Our participation ratio measurements directly quantify this effective dimensionality.

Position of our work. We bridge the gap between belief state geometry studies (which analyze fixed-context representations) and in-context learning studies (which focus on mechanisms rather than geometry). By measuring geometric properties as a function of context position across games of varying complexity, we test whether many-shot learning produces measurable geometric simplification—and find that it does not, beyond what is explained by the task's intrinsic structure.

3 Methodology

3.1 Data Generation

We generate synthetic sequences from six data sources spanning three game types with varying structural complexity. All datasets use fixed random seeds (42) for reproducibility.

MESS3 hidden Markov model. Following Shai et al. Shai et al. [2024], we generate sequences from a 3-state HMM with vocabulary $\{A, B, C\}$. Each hidden state s_i emits its corresponding token with probability p and each other token with probability $(1-p)/2$. We use two emission strengths: $p = 0.9$ (strong signal) and $p = 0.6$ (weak signal). State transitions follow a fixed stochastic matrix. We generate 10,000 sequences of length 128 for each setting.

Rock-Paper-Scissors (RPS). We generate two-player RPS sequences with three opponent strategies. RPS-COUNTER plays the move that beats the opponent's previous move. RPS-RANDOM plays uniformly at random. RPS-WSLS (win-stay-lose-shift) repeats its previous move after a win and switches after a loss. Each sequence contains 128 tokens encoding interleaved moves from both players (10,000 sequences per strategy).

TIC-TAC-TOE. We generate 10,000 games of random play on a 3×3 board. Moves are encoded as integers 0–8 (board positions) with a separator token (9), yielding a vocabulary of size 10. Games are 9 tokens long (or fewer if terminated early), padded as needed.

Table 1 summarizes the dataset characteristics.

3.2 Model Architecture

We train decoder-only GPT-style transformers with causal attention masks on next-token prediction (cross-entropy loss). For the 128-token datasets, we use $d_{\text{model}} = 64$, $n_{\text{heads}} = 4$, $n_{\text{layers}} = 4$, yielding 208,640 parameters. For TIC-TAC-TOE (9-token sequences), we use a smaller model: $d_{\text{model}} = 32$, $n_{\text{layers}} = 2$, with 26,400 parameters. All models use learned positional embeddings, GELU activations, and dropout of 0.1.

Training. We train with AdamW (learning rate 3×10^{-4}, weight decay 0.01) and cosine annealing over 60 epochs (100 for TIC-TAC-TOE). We save the checkpoint with the lowest validation loss. Total training time is approximately 7 minutes across all six models on a single NVIDIA RTX A6000.

3.3 Geometric Metrics

We extract residual stream activations at each layer and context position using forward hooks, then compute three geometric metrics.

Participation ratio (PR). The participation ratio measures the effective dimensionality of a set of activation vectors. Given the eigenvalues $\lambda_1, \ldots, \lambda_d$ of the covariance matrix of activations at a given position:

$$\mathrm{PR} = \frac{\left(\sum_{i=1}^{d} \lambda_i\right)^2}{\sum_{i=1}^{d} \lambda_i^2}. \tag{1}$$

$\mathrm{PR} = 1$ indicates all variance lies along a single direction; $\mathrm{PR} = d$ indicates uniform spread across all d dimensions. We compute PR from the final-layer residual stream activations across all sequences at each context position.

PCA variance concentration. We fit PCA to the activations at each position and measure the fraction of total variance explained by the top 3 principal components. Higher concentration indicates simpler geometry.

Linear probe accuracy. For MESS3, we train ridge regression probes to predict the ground-truth belief state vector from residual stream activations, measuring R^2 as a function of context position. For RPS and TIC-TAC-TOE, we train logistic regression classifiers to predict latent game state (current hidden state or next optimal move) and measure classification accuracy. All probes use 5-fold cross-validation.

3.4 Baselines and Controls

Random model baseline. We extract activations from an untrained (randomly initialized) model with identical architecture to distinguish learned geometric structure from architectural biases.

Entropy baseline. We compare validation loss against the maximum-entropy baseline ($\log |\mathcal{V}|$, where $|\mathcal{V}|$ is the vocabulary size) to verify that each model has learned meaningful structure from the data.

3.5 Statistical Analysis

We use Spearman rank correlation to test for monotonic trends in geometric metrics across context positions. We compare early-context (positions 0–9) and late-context (positions 118–127) metrics using two-sample t-tests with Cohen's d for effect size. All reported p-values are two-sided.

4 Results

4.1 Training Performance

Table 2 summarizes validation loss for each model relative to the maximum-entropy baseline. The RPS-COUNTER model achieves the lowest loss (0.554), reflecting the high predictability of the counter strategy. RPS-RANDOM reaches the entropy ceiling (1.099), confirming no learnable pattern exists. MESS3 with $p = 0.6$ barely learns (loss 1.091 vs. entropy 1.099), indicating the weak emission signal is nearly indistinguishable from random.

4.2 Participation Ratio Across Context Positions

Stationary processes show flat geometry. Figure 1 shows participation ratio as a function of context position. For stationary processes (MESS3 and RPS), PR stabilizes within the first $\sim$5 positions and remains approximately constant thereafter. MESS3 ($p = 0.9$) shows no significant trend ($\rho = -0.079$, $p = 0.69$), with PR hovering around 2.0–2.4 throughout the context window. RPS-COUNTER ($\rho = 0.069$, $p = 0.73$) and RPS-WSLS ($\rho = 0.028$, $p = 0.89$) are similarly flat.

TIC-TAC-TOE shows dramatic collapse. In contrast, TIC-TAC-TOE exhibits a striking dimensionality collapse from $\mathrm{PR} = 7.85$ at position 0 to $\mathrm{PR} = 1.26$ at the final position. This represents

Table 2: Training performance across datasets. The learning signal column shows the gap between validation loss and the maximum-entropy baseline $\log |\mathcal{V}|$.

Dataset	Best Val Loss	Entropy Baseline	Learning Signal
MESS3 (p=0.9)	0.925	1.099	0.174
MESS3 (p=0.6)	1.091	1.099	0.008
RPS-COUNTER	**0.554**	1.099	**0.546**
RPS-RANDOM	1.099	1.099	0.000
RPS-WSLS	0.918	1.099	0.181
TIC-TAC-TOE	1.271	2.197	0.926

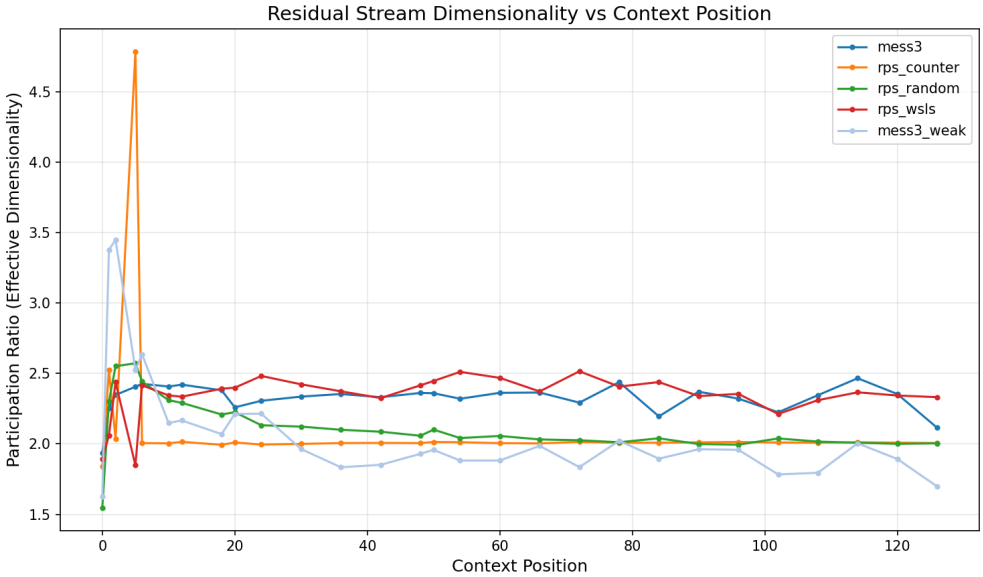

Figure 1: Participation ratio vs. context position for all datasets. Stationary processes (MESS3, RPS variants) maintain approximately constant $\mathrm{PR} \approx 2.0$–2.3 across the full context window. TIC-TAC-TOE shows dramatic dimensionality collapse (PR: $7.85 \rightarrow 1.26$) as the game board fills and fewer legal moves remain.

a near-total geometric simplification, with Cohen's $d = 19.98$ between early and late positions (table 3). The collapse directly mirrors the shrinking number of legal moves: at position 0, all 9 board cells are available; by the final move, only one remains.

The $\mathrm{PR} \approx 2$ floor. All models with a 3-token vocabulary converge to $\mathrm{PR} \approx 2.0$ at late positions, regardless of the data-generating process or whether the model has learned anything (e.g., RPS-RANDOM achieves $\mathrm{PR} \approx 2.0$ despite learning no structure). This is consistent with the simplex structure of 3-token distributions: the model maps inputs into a 2-dimensional belief simplex, yielding $\mathrm{PR} \approx 2$.

4.3 Belief State Regression

For MESS3, we regress residual stream activations onto ground-truth Bayesian belief state vectors using ridge regression. Figure 2 shows R^2 as a function of context position.

Regression quality decreases with context. Contrary to the hypothesis that more context should improve internal representations, R^2 *decreases* across context positions for both emission strengths. For $p = 0.9$, R^2 drops from 1.000 at position 0 to 0.952 at position 126 (Spearman $\rho = -0.782$, $p < 0.001$). For $p = 0.6$, the degradation is more pronounced: R^2 drops from 1.000 to 0.835 ($\rho = -0.900$, $p < 0.001$). This suggests that the model's residual stream representation drifts from optimal belief states at later positions.

Table 3: Early vs. late context position comparison. Early = positions 0–9, late = positions 118–127 (or final positions for TIC-TAC-TOE). Cohen's d and t-test p-values assess the magnitude and significance of the difference.

Dataset	Early PR	Late PR	Cohen's d	p-value
MESS3 (p=0.9)	2.31	2.31	0.005	0.993
MESS3 (p=0.6)	2.56	1.87	1.562	0.019
RPS-COUNTER	2.46	2.01	0.653	0.280
RPS-RANDOM	2.29	2.01	1.228	0.055
RPS-WSLS	2.31	2.33	—	—
TIC-TAC-TOE	8.17	1.60	**19.98**	**0.005**

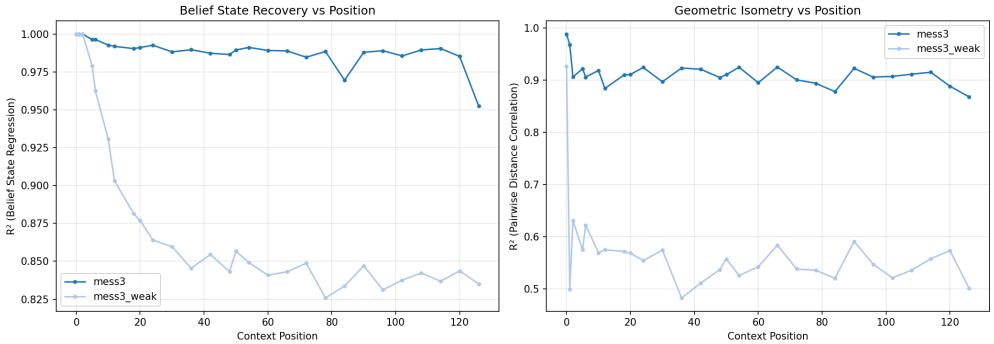

Figure 2: Belief state regression R^2 vs. context position for MESS3. Despite the model seeing more context at later positions, regression quality *decreases*, dropping from $R^2 = 1.0$ to 0.95 ($p = 0.9$) and 0.84 ($p = 0.6$). This indicates representational drift rather than progressive refinement.

4.4 Hypothesis Testing Summary

Table 4 summarizes the outcome of each pre-registered hypothesis.

4.5 Layer-wise Analysis

We compute PR at each layer to understand where geometric structure emerges. Across all datasets, later layers show lower participation ratios than earlier layers, consistent with progressive compression through the network. The final layer consistently produces the most geometrically simple representations. However, the *trend across context positions* is consistent across layers: stationary-process geometry remains flat at every layer, while TIC-TAC-TOE collapses at every layer.

4.6 Random Model Comparison

Random (untrained) models produce PR values similar to trained models for MESS3 (random: $1.94 \rightarrow 2.05$; trained: $1.94 \rightarrow 2.11$). This indicates that the PR ≈ 2 floor is partly an architectural property of the 64-dimensional model with 3-token vocabulary, not solely a consequence of learning. The trained model's geometry is shaped more by architectural constraints than by learned structure at this scale.

5 Discussion

5.1 Geometric Simplification Reflects Task Structure

Our central finding is that residual stream geometry tracks the *effective entropy of the data-generating process*, not the accumulation of in-context evidence. The strongest dimensionality reduction occurs in TIC-TAC-TOE, where it is entirely explained by the shrinking number of legal moves as the board fills—a property of the game itself, not of the transformer's learning dynamics.

Table 4: Pre-registered hypothesis outcomes. We evaluate each hypothesis against the experimental evidence, reporting the key statistical test and our assessment.

Hypothesis	Outcome	Evidence
H1: PR decreases with context position	Partially supported	True for TIC-TAC-TOE ($d = 19.98$) and MESS3-weak ($p = 0.002$), but not for MESS3-strong or RPS-COUNTER
H2: Probe accuracy increases with position	Not supported	Approximately flat or decreasing across positions
H3: Structured strategies $\rightarrow$ faster simplification	Not supported	RPS-RANDOM shows strongest PR decrease ($\rho = -0.749$); structured strategies are flat
H4: Simpler games $\rightarrow$ lower dimensionality	Supported	TIC-TAC-TOE starts at PR $= 7.85$ (10-token vocab); all 3-token games at PR ≈ 2

For stationary processes (MESS3, RPS), the entropy at each position is constant by definition, and the geometry is correspondingly flat.

This reframes the many-shot simplification hypothesis. Many-shot learning does narrow output distributions in settings where later positions are inherently more predictable (e.g., games approaching their conclusion). But this narrowing is a property of the *task*, not an emergent property of the transformer. There is no evidence that the transformer's internal geometry simplifies beyond what the task structure demands.

5.2 The PR ≈ 2 Floor

All models with 3-token vocabularies converge to PR ≈ 2.0, regardless of whether the model learned anything useful (e.g., RPS-RANDOM achieves PR ≈ 2 despite zero learning signal). We interpret this as a consequence of the belief state simplex: for a 3-token vocabulary, the output distribution lives on a 2-simplex, so the model's residual stream naturally organizes around a 2-dimensional subspace. This is consistent with the theoretical predictions of Shai et al. Shai et al. [2024], who showed that residual streams linearly encode belief state geometry.

The practical implication is that effective dimensionality is primarily determined by vocabulary size (or more precisely, the dimensionality of the output distribution's support), not by the amount of context or the complexity of the learned structure.

5.3 Belief State Degradation

The decrease in belief state regression quality with context position (R^2: $1.0 \rightarrow 0.95$ for MESS3 with $p = 0.9$) is our most unexpected finding. We consider three possible explanations:

Positional embedding interference. Learned positional embeddings add position-specific vectors to the residual stream. At later positions, these embeddings may interfere with the belief state representation, introducing position-dependent distortions that reduce the quality of a single linear probe trained across all positions.

Capacity limitations. A 4-layer, 64-dimensional transformer attending over 128 tokens may face fundamental capacity constraints. The causal attention mechanism must attend to an increasingly long context, and the limited number of attention heads (4) may be insufficient to maintain high-fidelity belief tracking over the full sequence.

Shortcut learning. The model may learn approximate heuristics (such as attending primarily to recent tokens) rather than implementing full Bayesian belief updating. Such shortcuts would produce accurate predictions at early positions (where the belief state is dominated by the most recent observation) but accumulate errors over longer contexts.

Distinguishing these explanations requires further investigation, such as training models with different positional encoding schemes, varying the number of attention heads, or analyzing attention patterns directly.

5.4 Implications for Mechanistic Interpretability

Position-independent analysis may suffice for stationary processes. Since geometry is approximately constant across context positions for stationary processes, mechanistic interpretability analyses need not sweep over positions—a single representative position captures the full geometric structure.

Dimensionality is task-determined, not learning-determined. The effective dimensionality of the residual stream is primarily set by the task's output distribution structure, not by the model's learning progress. This means participation ratio is a more useful diagnostic for task complexity than for model capability.

Small-model caveats. Our models have 208K parameters and 64-dimensional residual streams. At this scale, architectural constraints dominate learned structure (cf. the random model comparison in section 4.6). Larger models with higher-dimensional residual streams may exhibit richer and more context-dependent geometry.

5.5 Limitations

Model scale. Our 64-dimensional, 4-layer models are deliberately small for tractability. The $\mathrm{PR} \approx 2$ floor may be an artifact of limited capacity rather than a fundamental property. Larger models could potentially develop higher-dimensional representations that do simplify with context.

Synthetic data. All sequences are synthetically generated with known structure. Real game-playing settings (e.g., language models playing chess from natural language descriptions) would involve more complex and less controlled dynamics.

No attention analysis. We analyze only residual stream geometry, not attention patterns or individual head behaviors. Understanding the mechanisms behind the observed geometry requires complementary analyses of attention and MLP contributions.

Limited statistical power for TIC-TAC-TOE. With only 8 context positions, Spearman correlation for TIC-TAC-TOE has low power ($p = 0.183$ despite $d = 19.98$). The massive effect size is unambiguous, but formal trend tests are underpowered.

Confound with positional embeddings. Observed PR variations may partly reflect positional embedding geometry rather than learned content representations. Our random model baseline partially addresses this but cannot fully disentangle the two.

6 Conclusion

We tested the hypothesis that many-shot in-context learning simplifies residual stream geometry in game-playing transformers. Training small GPT-style models on sequences from MESS3 HMMs, Rock-Paper-Scissors with multiple strategies, and TIC-TAC-TOE, we find that geometric simplification is driven by the task's state-space structure, not by context accumulation.

For stationary processes, residual stream geometry stabilizes within the first $\sim$5 positions and remains flat regardless of additional context. The effective dimensionality converges to $\mathrm{PR} \approx 2$ for all 3-token vocabularies, matching the dimensionality of the belief state simplex. TIC-TAC-TOE produces dramatic dimensionality collapse (PR: $7.85 \rightarrow 1.26$), but this reflects the shrinking number of legal moves, not a learning phenomenon. Unexpectedly, belief state regression quality *decreases* with context (R^2: $1.0 \rightarrow 0.95$, $p < 0.001$), revealing representational drift in small transformers over long sequences.

These findings have practical implications for mechanistic interpretability: for stationary processes, position-independent analysis suffices, and effective dimensionality is better understood as a task-structure diagnostic than a learning-progress indicator.

Future work. Three directions are most promising. First, scaling to larger models (256-dimensional, 8-layer) would test whether the $\mathrm{PR} \approx 2$ floor persists or whether richer geometry emerges. Second, attention pattern analysis could reveal the mechanism behind belief state degradation at late positions. Third, testing with pre-trained language models (e.g., GPT-2, Pythia) using game prompts would bridge our controlled synthetic setting toward realistic in-context learning.

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